

Tech For Tomorrow 2026

Krishi AI

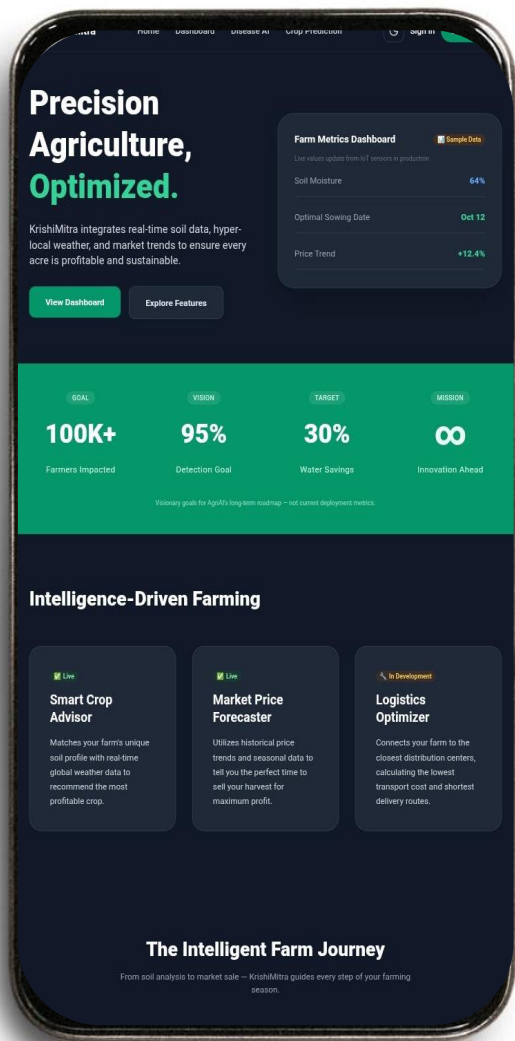
Problem statement

- "From disease outbreaks to weather uncertainty and fluctuating market prices, farmers must make multiple high-impact decisions with limited access to real-time, localized insights."
- **Problem Statement Title:** Krishi AI: Bridging Technology and Agriculture
- **Theme:** Smart Farming for a Better Future
- **PS Category:** Software
- **Team Name:** QuantumCoders



IDEA TITLE

Empowering Farmers with AI



DETAILED EXPLANATION

- AI-based crop recommendation using soil & weather.
- Crop disease detection through image analysis.
- Crop price prediction using AI.
- Multilingual voice assistant and website.
- One integrated platform for smart farming.

HOW IT ADDRESSES THE PROBLEM

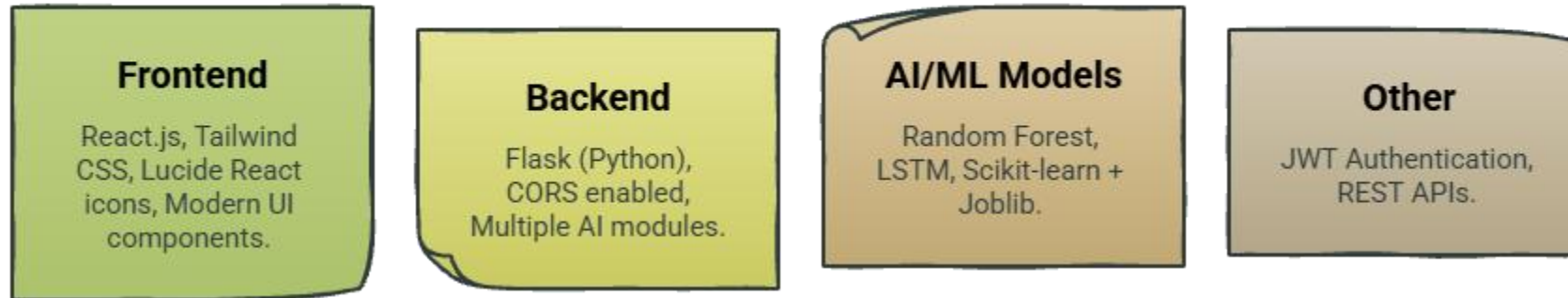
- Helps farmers choose the right crop.
- Detects diseases early to reduce losses.
- Protects crops with weather alerts.
- Predicts better selling prices.
- Breaks language barriers with voice support.

INNOVATION & UNIQUENESS

- All-in-one smart farming platform.
- AI-powered crop recommendations.
- Real-time weather and risk alerts.
- Scalable platform with future IoT integration.

TECHNICAL APPROACH

Technologies used



Core Features

Feature	 Crop Recommendation	 Market Price Prediction	 Fertilizer Recommendation	 Smart Irrigation Advisor	 Dashboard
Status	Live	Live	Live	Live	Live
Description	AI suggests best crop	LSTM model forecasts mandi prices	Gives chemical + organic options	Hyper-local drip irrigation advice	Real-time farm metrics

FEASIBILITY AND VIABILITY



FEASIBILITY ANALYSIS

- Uses existing technologies like React.js ,Tailwind CSS,Lucide React icons,Flask (Python)...
- Public datasets for crop, weather, and diseases are already available.
- Weather APIs and satellite data can be integrated easily.
- Voice assistant and multilingual APIs are readily available.
- Mobile and web platforms ensure accessibility.



POTENTIAL CHALLENGES & RISKS

- Low internet connectivity in rural areas.
- Inaccurate or incomplete agricultural datasets.
- Farmers may have limited digital literacy.
- Integration of multiple APIs may create compatibility issues.
- AI predictions may not always be fully accurate.



STRATEGIES TO OVERCOME THESE CHALLENGES

- Conduct awareness programs and farmer training sessions.
- Use cloud services and government partnerships to reduce cost.
- Add multilingual voice guidance for easy usage.
- Use verified and multiple agricultural datasets for training.
- Provide offline support and lightweight application features.

IMPACT AND BENEFITS



IMPACT ON TARGET AUDIENCE

- Better crop selection through AI recommendations
- Early crop disease detection
- Improved decision-making using weather alerts
- Easy access through multilingual support
- Increased productivity and efficiency



SOCIAL IMPACT

- Promotes digital inclusion in rural areas
- Improves quality of life for farmers
- Improves access to agricultural knowledge
- Encourages smart farming adoption
- Strengthens rural communities



ECONOMIC IMPACT

- Increases farmer income
- Reduces crop losses
- Optimizes resource usage and cost
- Eliminates middlemen dependency
- Supports sustainable economic growth



RESEARCH AND REFERENCES

Features	Krishi AI	kisan Suvidha	plantix	AgriApp	DeHaat
AI Crop Recommendation	✓	✗	✗	Limited	Limited
Market Price Prediction (LSTM)	✓	Shows prices only	✗	✗	Limited
Smart Irrigation Advisor	✓ Hyper-local AI advice	✗	✗	Limited	✗
Crop Disease Detection	✓	✗	✓	Limited	✗

